Technical Skills Overview

Programming Fundamentals

- Use of conditional statements and loops (if, switch, for, while) in C++, C#, Java
- Working with basic data structures (lists, dictionaries) in C++, C#, Java
- Writing clean and readable code (proper naming conventions, method decomposition)
- Basic exception and error handling (try–catch)

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Object-Oriented Programming

- Object-oriented programming concepts (inheritance, polymorphism)
- Understanding and practical use of classes and objects
- Basic code organization using object-oriented principles (C++, C#, Java)

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Databases & SQL (MySQL)

- Designing simple relational databases
- One-to-one, one-to-many and many-to-many relationships
- Use of primary keys and foreign keys
- Creating and modifying database structures (CREATE, ALTER, DROP)
- Writing SQL queries:

SELECT, JOIN, GROUP BY, ORDER BY, INSERT, UPDATE, DELETE

Web Development

- Creating HTML pages using forms, tables and CSS
- Separation of concerns: HTML / CSS / JavaScript / PHP
- Creating HTML forms with basic validation
- Implementing basic JavaScript functionality
- Handling simple events (onclick, onchange)
- Displaying database content using PHP
- Connecting PHP with MySQL (mysqli / PDO)

Development Tools

- Using Integrated Development Environments (IDEs): IntelliJ IDEA, Visual Studio, Visual Studio Code
- Basic use of version control systems (Git)

Graphics & Multimedia

- Basic image editing using GIMP
- Creating simple animations in GIMP

Server-Side JavaScript (Node.js) (introductory level)

- Basic knowledge of the Node.js runtime environment
- Understanding the role of JavaScript outside the browser
- Creating simple console-based and server-side scripts
- Working with basic modules and dependencies
- Understanding the concept of asynchronous execution

Desktop Application Development (Windows Forms)

- Basic knowledge of Windows Forms application structure
- Creating simple graphical user interfaces
- Handling basic user interactions (buttons, input fields)
- Separating user interface and application logic

(C# / .NET environment)

Mobile Application Development (Android Studio) (introductory level)

- Basic familiarity with the Android Studio development environment
- Creating simple mobile application layouts
- Understanding basic application structure and lifecycle concepts
- Handling simple user interactions

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